

Directional Perturbation of Image Covariance Geometry: Compression Response Across the Kodak Lossless True Color Suite

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Abstract

Previous work established that the 24 images of the Kodak Lossless True Color Image Suite span a broad range of image-specific covariance geometries, that the BT.601 fixed transform incompletely decorrelates those structures, and that chrominance-axis misalignment is constrained to a single effective degree of freedom. This paper applies directional perturbations to image covariance structure, operating in RGB pixel space upstream of any colorspace conversion, and measures the resulting geometric and compression response. For each image, three independent perturbations — targeting the red, green, and blue channel axes respectively — were applied to the covariance structure, displacing all 72 resulting images from the chrominance constraint manifold with direction-dependent geometric signatures observed across the suite. Through a third-party compression pipeline (Facebook/Meta), perturbed images produce a mean bits-per-pixel reduction of 54.5% relative to originals (range 29.1%–86.6%), while a control condition applying the same pipeline without covariance perturbation produces no comparable change. The geometric displacement persists through seven rounds of lossy JPEG recompression, retaining 84–97% of the initial displacement.

1 Established Geometry

The Kodak Lossless True Color Image Suite comprises 24 images spanning a broad range of inter-channel redundancy structures. Per-image eigendecomposition of the 3×3 RGB covariance matrix yields condition numbers from 7.55 to 1739.16, corresponding to five distinct dimensionality tiers — from near-spherical three-channel distributions to extremely elongated, near-degenerate structures [1]. When measured against per-image KLT ceilings, the BT.601 fixed transform achieves between 31.5% and 82.3% of optimal decorrelation, leaving a mean Cb–Cr residual correlation of 0.616 across the suite [2]. Angular misalignment between each image’s eigenvectors and BT.601’s fixed axes accounts for 57.6% of the variance in decorrelation efficiency, and the two chrominance misalignment angles θ_2 and θ_3 co-vary at $r = 0.9993$, constraining the chrominance misalignment space to a single effective degree of freedom [3, 4].

Table 1. Inherited geometry from [1]–[4].

Metric	Value	Source
Condition number range	7.55–1739.16	[1]
BT.601 efficiency range	31.5%–82.3%	[2]
Mean residual Cb–Cr correlation	0.616	[2]
θ_2 – θ_3 correlation	0.9993	[4]
Misalignment–efficiency R^2	0.576	[3]

2 Intervention

The measurements reported in [1]–[4] are observational. They characterize existing covariance structure and its alignment with a fixed transform but do not alter either. The present work applies directional perturbations to the covariance structure itself, operating in RGB pixel space upstream of any colorspace conversion, and measures the geometric and compression response.

For each of the 24 Kodak images, three independent perturbations were applied, each targeting a single channel axis: red, green, or blue. Each perturbation reshapes the covariance matrix rather than rotating it, producing inter-channel correlation structures not obtainable through fixed orthonormal rotation of the original RGB basis. The 72 resulting images are standard files — JPEG, TIFF, or PNG — readable by any decoder without modification.

A red-targeted perturbation reduces correlation between the red channel and both remaining channels ($R-G < 0.30$, $R-B < 0.30$) while preserving coupling within the non-targeted pair ($G-B > 0.90$). Green- and blue-targeted perturbations follow the analogous pattern. In all cases, the target channel’s standard deviation was required to remain above 10 to prevent channel collapse. All 72 perturbations satisfied these thresholds without exception.

Table 2. Perturbation protocol.

Perturbation	Target constraints	Preserved pair	Channel std
Red	$R-G < 0.30$, $R-B < 0.30$	$G-B > 0.90$	$R \text{ std} > 10$
Green	$R-G < 0.30$, $G-B < 0.30$	$R-B > 0.90$	$G \text{ std} > 10$
Blue	$R-B < 0.30$, $G-B < 0.30$	$R-G > 0.90$	$B \text{ std} > 10$

n = 72 (24 images × 3 directions). All 72 satisfied thresholds without exception.

3 Geometric Response

The 24 original Kodak images occupy a narrow region of θ_2 – θ_3 space, lying along the regression line $\theta_3 = 1.004 \cdot \theta_2 - 2.372$ established in [4]. Under directional perturbation, all 72 redistributed versions depart from this manifold, and the direction of departure corresponds to the targeted channel axis.

Red-targeted perturbations shift θ_2 and θ_3 while approximately preserving their linear relationship, producing displacements that remain near or loosely parallel to the original constraint line. Green-targeted perturbations displace θ_3 downward relative to θ_2 , moving images below the constraint line with a shallower apparent slope. Blue-targeted perturbations produce the largest departures: θ_3 drops to a suite mean of 17.96° (from an original suite mean of 47.11°), while θ_2 shifts moderately, yielding positions far below the constraint manifold.

These displacement patterns appear across all dimensionality tiers, condition numbers, and baseline correlation structures represented in the suite. The geometric signatures are also format-independent: the same perturbation measured in JPEG, TIFF, and PNG produces mean Avg $|r|$ deltas below 0.001 across all 72 versions.

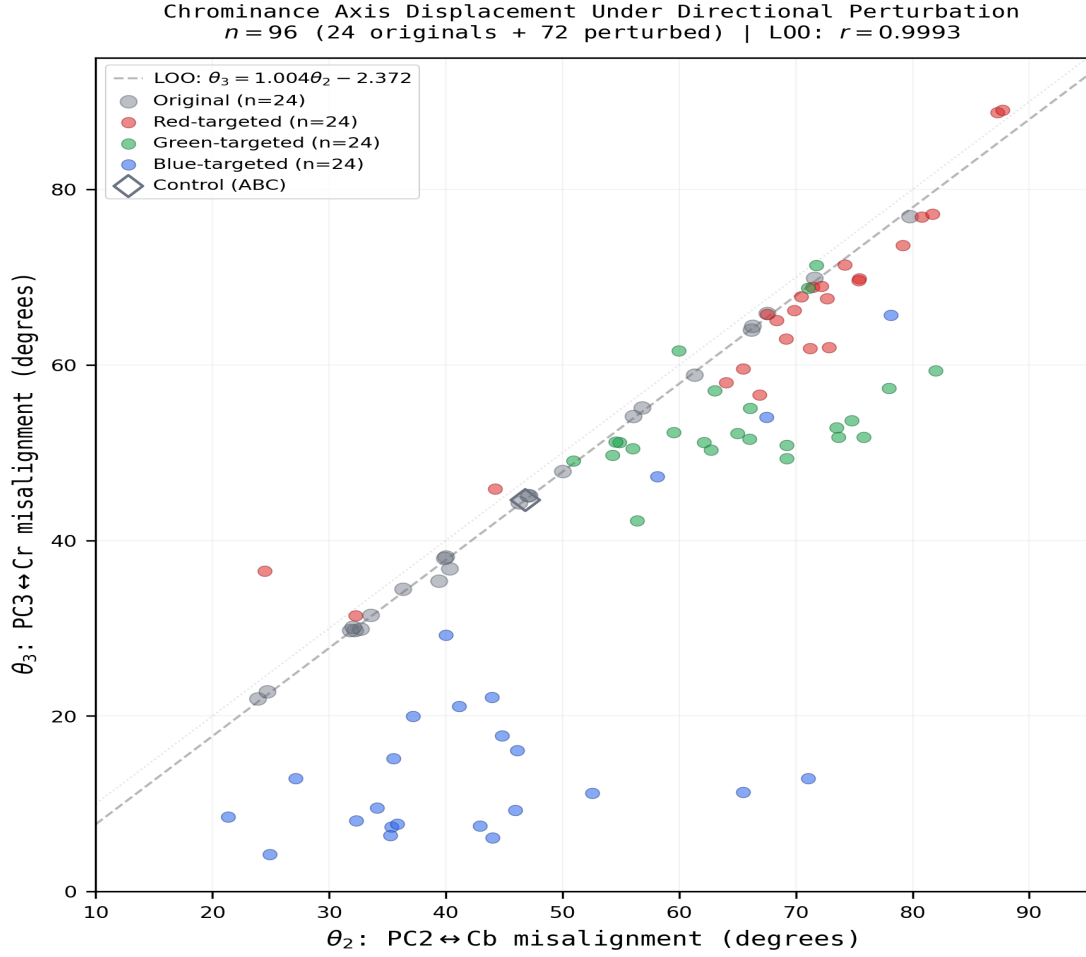


Figure 1. Chrominance axis displacement under directional perturbation. Gray dots: 24 original Kodak images on the LOO constraint line ($\theta_3 = 1.004\theta_2 - 2.372$, $r = 0.9993$). Colored dots: 72 perturbed versions (24 per direction). Diamond: ABC control (no perturbation). Blue-targeted perturbations produce the largest manifold departures.

Table 3. Geometric displacement summary (2560 JPG SPDR).

Direction	n	Mean θ_2	Mean θ_3	Mean LOO dev	Mean $ r $
Original	24	46.75°	44.56°	0.00°	0.848
Red	24	68.52°	65.06°	-1.37°	0.411
Green	24	65.43°	53.85°	-9.47°	0.407
Blue	24	44.19°	17.96°	-24.03°	0.388
Control (ABC)	24	46.81°	44.61°	-0.01°	0.846

Source: 768 OG PNG (originals), 2560 JPG SPDR (perturbations), ABC no-FB (control).

A control condition was measured in which images were processed through the identical pipeline — including resolution change and format conversion — without covariance perturbation. The control images preserved their original correlation structure (Avg $|r| = 0.893$ vs. 0.898), remained on the LOO constraint line (deviation $< 0.5^\circ$), and maintained their original dimensionality classification. The geometric displacement observed in the 72 perturbed versions is attributable to the covariance perturbation, not to the pipeline.

4 Compression Response

The geometric displacement documented above corresponds to a measurable change in compression behavior. Each of the 24 images was uploaded to Facebook (Meta) in both its original form and its three perturbed forms. The platform re-encodes all uploads as JPEG; the compressed outputs were downloaded and measured.

Across the full Facebook pipeline, perturbed images produced a mean BPP reduction of 54.5% relative to originals at the platform’s output resolution of 2048×1366. Reductions ranged from 29.1% to 86.6%. All 72 perturbed versions produced lower output BPP than their corresponding originals through the tested pipeline.

The perturbed images also carry a higher pixel count than the originals (2048×1366 vs. 768×512 at Facebook’s output resolution, a 7.1× ratio). The BPP reduction is computed at the output resolution, not the input, and therefore reflects compression efficiency independent of the resolution difference.

To isolate the quality-setting response, each image was exported at JPEG quality levels Q60, Q75, and Q90 in both original and perturbed form, then uploaded through the Facebook pipeline. For originals, the Q60-to-Q90 spread in output BPP averaged 1.586 across the suite. For perturbed versions, the corresponding spread averaged 0.002.

In all 24 images, perturbed versions exported at Q60 produced lower output BPP than originals exported at Q90 through the same third-party pipeline, at 7.1× the pixel count.

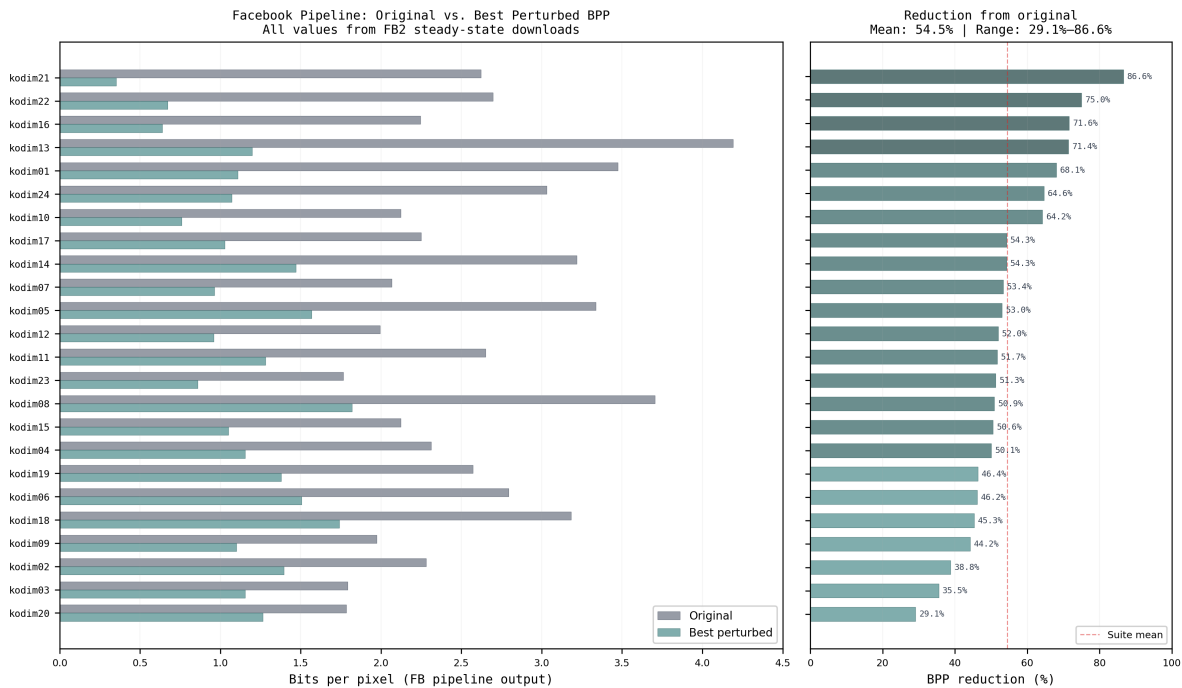


Figure 2. Facebook pipeline compression response. Left: original vs. best perturbed BPP for all 24 images. Right: per-image BPP reduction percentage. Dashed line: suite mean (54.5%). All values from FB2 steady-state downloads, verified May 7, 2026.

Table 4. Compression response summary (Facebook pipeline).

Metric	Original	Perturbed
Mean BPP (FB output)	2.591	1.366
Q60–Q90 spread (FB output)	1.586	0.002
Mean BPP reduction	—	54.5%
Min BPP reduction	—	29.1%
Max BPP reduction	—	86.6%

Pipeline: OG = 768×512 PNG → FB → 768×512 JPEG. Perturbed = 2560×1707 JPG → FB → 2048×1366 JPEG. All values from FB2 (steady state). 72/72 perturbed versions below OG baseline. Best single: kodim21 BLU at 0.351 BPP (86.6% reduction).

5 Structural Permanence

To test whether the perturbation persists under repeated lossy encoding, perturbed images were passed through up to seven successive rounds of JPEG recompression, with geometric measurements taken at each pass.

On the first compression pass, the perturbed version of kodim01 (blue-targeted) shifted from the original LOO deviation of $+0.46^\circ$ to -15.37° . Through seven subsequent rounds of lossy recompression, the deviation converged toward approximately -13° , retaining 84–97% of the initial geometric displacement. The correlation drift between successive passes decreased monotonically. File size stabilized at 142 KB after pass 2.

The geometric displacement remains measurable through repeated lossy recompression.

Table 5. Structural permanence (kodim01, blue-targeted).

Pass	LOO deviation	Retained displacement	File size
Original	$+0.46^\circ$	—	719 KB
1×	-15.37°	—	144 KB
2×	-14.26°	90%	142 KB
4×	-13.06°	85%	142 KB
7×	-12.66°	84%	142 KB

Retained displacement = percentage of pass-1 LOO shift preserved at each subsequent pass.

6 Discussion

The measurements presented here extend the observational framework of [1]–[4] into an interventional one. Where the preceding papers characterize covariance geometry and its relationship to fixed-transform decorrelation, the present work alters that geometry directly and measures the downstream response across two independent surfaces: angular displacement in the θ_2 – θ_3 misalignment space, and bits-per-pixel behavior through third-party compression pipelines.

The control condition constrains the operative variable. Resolution change and format conversion alone do not displace images from the LOO constraint manifold, do not alter the dimensionality classification, and do not produce comparable BPP reduction. Comparable compression response was not observed in the absence of covariance perturbation.

Three properties of the perturbation are consistently observed across the measurements. First, it operates upstream of any colorspace conversion or encoder, on standard RGB pixel data, and its output is readable by existing decoders without modification. Second, the geometric displacement it produces persists through multiple rounds of lossy recompression. Third, three independent perturbation directions applied to each image produce three distinct geometric signatures across the full range of covariance structures in the suite.

The scope of these findings is bounded by the dataset. All measurements were performed on the 24 images of the Kodak Lossless True Color Image Suite, a standard benchmark but a limited sample of natural image statistics. The compression measurements reported here were obtained under a specific pipeline configuration; the interaction between perturbation parameters and pipeline conditions was not exhaustively characterized. Whether the observed relationships between covariance perturbation and compression response generalize to larger or more diverse image populations is an open empirical question. The measurement methodology is identical to that used in [1]–[4];

complete per-image data is provided as machine-readable JSON files for independent verification.

References

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